

ABRAR TAHER

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EDUCATION

M.Sc. in Computer Science and Engineering

Chittagong University of Engineering & Technology

2024 – 2026

CGPA: 3.83/4.00

Thesis Pending

B.Sc. in Computer Science and Engineering

Chittagong University of Engineering & Technology

2018 – 2023

CGPA: 3.37/4.00

Last Four Semesters CGPA: 3.78/4.00

RESEARCH INTEREST

Multimodal Information Retrieval, AI in Healthcare, Medical Image Processing, Computer Vision, Explainable AI, Deep Learning, Machine Learning

RESEARCH EXPERIENCE

Undergraduate Thesis: A Deep Learning Approach for Multiclass Brain Tumor Classification and Segmentation

Supervisor: Ms. Sabiha Anan, Assistant Professor, Department of Computer Science and Engineering

- Developed a custom Convolutional Neural Network for multiclass brain tumor classification and a Residual Attention U-Net architecture for precise tumor segmentation.
- Conducted performance benchmarking against pretrained models while gaining hands-on experience with T1-weighted MR image preprocessing, feature extraction for small-scale tumors.

ONGOING WORK

Master's Thesis: Pathology Preserving Lesion–Vessel Interaction Learning for Progression-Aware Diabetic Retinopathy Grading

- Retinal biomarker characteristics and lesion–vessel structural relationships to capture clinically relevant pathological and vascular information associated with different stages of diabetic retinopathy.
- Vascular abnormalities with progression-aware ordinal learning to model the ordered transition of disease severity and improve discrimination between adjacent and advanced DR stages

PUBLICATIONS

- "TrustKD-LULC: Reliable Knowledge Distillation for Remote Sensing Image Classification", ICECER 2026 (Accepted)
- S. Saikat, **A. Taher** et al., "An Explainable Ensemble Framework for Multi-Level Diabetic Retinopathy Classification", 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence Networking (QPAIN). DOI: [10.13140/RG.2.2.27830.05443](#)
- S. Mallick, A. Debnath, **A. Taher** et al., "Early Detection of Diabetic Retinopathy using Novel Pre-Processing of Retinal Fundus Images", 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence Networking (QPAIN). DOI: [10.13140/RG.2.2.27410.62400](#)
- A. Debnath, S. Mallick, **A. Taher** et al., "Gamma-CLAHE Enhanced InceptionV3 for Alzheimer's and Parkinson's Disease Detection with Explainable MRI Insights", 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence Networking (QPAIN). DOI: [10.13140/RG.2.2.32863.21920](#)

- **A. Taher** and S. Anan, "Multiclass Brain Tumor Classification and Segmentation from 2D MR Images: A Deep Learning Approach Using Custom CNN and Residual Attention U-Net," 26th International Conference on Computer and Information Technology (ICCIT), pp. 1-6, 2023. DOI: [10.1109/ICCIT60459.2023.10441606](https://doi.org/10.1109/ICCIT60459.2023.10441606)
- **A. Taher**, W. I. Z. Ayon, and M. S. Hossain, "Histopathological Image-Based Classification of Lung and Colon Cancer Using Deep Learning Architectures with Preprocessing Enhancements," in Proceedings of the 27th International Conference on Computer and Information Technology (ICCIT), 2024. DOI: [10.1109/ICCIT64611.2024.11022478](https://doi.org/10.1109/ICCIT64611.2024.11022478)
- S. Mallick, A. Debnath, **A. Taher**, and A. Mozumder, "BreaKNet: Fusing Depthwise Convolutional and Shifted-Window Attention Features for Multiclass Breast Tumor Subtype Classification on Histopathological Images" in Proceedings of the 2026 6th International Conference on Electrical, Computer and Energy Technologies (ICECET) DOI
- **A. Taher** and W. I. Z. Ayon, "Exploring Sleep Disorders: A Comparative Analysis of Machine Learning Algorithms on Sleep Health and Lifestyle Data," 2024 IEEE International Conference on Power, Electrical, Electronics, and Industrial Applications (PEEIACON-24), 2024. DOI: [10.1109/PEEIACON63629.2024.10800593](https://doi.org/10.1109/PEEIACON63629.2024.10800593)

SKILLS

Programming Language	Python, C, PHP, MySQL
Frameworks & Libraries	Keras, TensorFlow, PyTorch, Scikit-learn, Pandas, Seaborn, Matplotlib
Paradigms	Algorithm Design, Statistical Modeling
Tools	Lucidchart, Draw.io, Power BI, Colab, Jupyter, Kaggle, Git/GitHub, Overleaf
Web Development	HTML5, CSS3, Node.js

WORK EXPERIENCE

Junior Instructor Computer Science Asian University For Women	January 14, 2024 - Present Chittagong, Bangladesh
• Course: Harvard CS-50, Programming With Python, Computational Thinking & Programming	

Lecturer Department of CSE Port City International University	July 04, 2023 – January 03, 2024 Chittagong, Bangladesh
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LANGUAGE PROFICIENCY TESTS

- **IELTS** – Overall Band Score: **7** (Listening: 8.0, Reading: 6.5, Writing: 6.5, Speaking: 6.5)

ACHIEVEMENTS

- Awarded the Bangladesh Technical Education Board Scholarship (BTEB) based on term results
- Received a merit based scholarship for the University Undergraduate Admission Test-2017
- Champion of the Regional Astronomy Olympiad (2014)
- Silver Medalist in the Regional Physics Olympiad (2013)

EXTRA-CURRICULAR ACTIVITIES

- **Community Service:** Volunteering at a village orphanage.
- **Athletics:** Participated in short-distance marathons promoting health and awareness.
- **Fitness:** Regular practice of fitness exercises and calisthenics training for personal well-being.
- **Fundraising:** Volunteered in fund raising and relief committee for the flood affected area in 2024